許樂瑋 112700026

16. The Cost of Lumber, continued The analysis of variance F-test in Exercise 15 (Section 11.2) indicated that there was a difference in the average cost of lumber for the four states. The following information from that exercise is given in the table:

Sample Means	$\bar{x}_1 = 262.2$	MSE	41.25
	$\bar{x}_2 = 234.8$	Error df:	16
	$\bar{x}_3 = 251.6$	n_i:	5
	$\bar{x}_4 = 268.6$	k:	4

Use Tukey's method for paired comparisons to determine which means differ significantly from the others at the $\alpha = .01$ level.

$\alpha = 0.01$

$k = 4$

error $df = 16$

$q_{0.01(4,16)} \approx 5.19$

$$W = q_{\alpha(k,df)} \sqrt{\frac{MSE}{n}}$$
$$= 5.19 \sqrt{\frac{41.25}{5}}$$
$$= 14.91$$

$|\bar{x}_1 - \bar{x}_2| = 262.2 - 234.8 = 27.4 > 14.91$, significant difference

$|\bar{x}_1 - \bar{x}_3| = 262.2 - 251.6 = 10.6 < 14.91$, Not significant difference

$|\bar{x}_1 - \bar{x}_4| = 262.2 - 268.6 = 6.4 < 14.91$, Not significant difference

$|\bar{x}_2 - \bar{x}_3| = 234.8 - 251.6 = 16.8 > 14.91$, significant difference

$|\bar{x}_2 - \bar{x}_4| = 234.8 - 268.6 = 33.8 > 14.91$, significant difference

$|\bar{x}_3 - \bar{x}_4| = 251.6 - 268.6 = 17.0 > 14.91$, significant difference

$\therefore$ $(1,2)$, $(2,3)$, $(2,4)$, $(3,4)$ significant difference from the others at the $\alpha = 0.01$ level.

DATA SET DS1113

17. GRE Scores The quantitative reasoning scores on the Graduate Record Examination (GRE)[3] were recorded for students admitted to three different graduate programs at a local university.

Graduate Program					
Life Sciences		Physical Sciences		Social Sciences	
630	660	660	760	440	540
640	660	640	670	330	450
470	480	720	700	670	570
600	650	690	710	570	530
580	710	530	450	590	630

a. Do these data provide sufficient evidence to indicate a difference in the mean GRE scores for applicants admitted to the three programs?

b. Find a 95% confidence interval for the difference in mean GRE scores for Life Sciences and Physical Sciences.

c. If you find a significant difference in the average GRE scores for the three programs, use Tukey's method for paired comparisons to determine which means differ significantly from the others. Use $\alpha = .05$.

LS: $x_1 = 6080$, $n_1 = 10$, $\bar{x}_1 = 608$

PS: $x_2 = 6530$, $n_2 = 10$, $\bar{x}_2 = 653$

SS: $x_3 = 5320$, $n_3 = 10$, $\bar{x}_3 = 532$

$N = 30$ $k = 3$

$\bar{x} = 597.67$

$x_{..} = 17930$

(a) $H_0: \mu_1 = \mu_2 = \mu_3$

$H_a: \mu_1, \mu_2, \mu_3$ 不完全相等

ANOVA

Source	SS	df	MS	F
Treatment	74806.67	$k-1 = 2$	$\frac{74806.67}{2} = 37403.34$	$\frac{37403.34}{8367.78} = 4.47$
Error	225930	$N-k = 27$	$\frac{225930}{27} = 8367.78$	
Total	300736.67	$N-1 = 29$		

$$CM = \frac{x_{..}^2}{N} = \frac{17930^2}{30} = 10716163.33$$

$$SSTR = \sum_{i=1}^{3} \frac{x_{i.}^2}{n} - CM = 10790970 - 10716163.33 = 74806.67$$

$$SSTO = \sum_i \sum_j x_{ij}^2 - CM = 11016900 - 10716163.33 = 300736.67$$

$$SSE = SSTO - SSTR = 225930$$

Test:

$F = 4.47$

$\because RR = \{F > f_{2,27,0.05} = 3.35\}$

$\therefore F \in RR$, reject H_0

(b) $\mu_1 - \mu_2$ 的 95% C.I. $= (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, N-k} \cdot \sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

$= (608 - 653) \pm t_{0.025, 27} \cdot \sqrt{8367.78\left(\frac{1}{10} + \frac{1}{10}\right)}$

$= -45 \pm 2.0518\ (40.909)$

$= -45 \pm 83.94$

$= (-128.94, 38.94)$

∵ C.I. 包含 0

∴ Life Science 和 Physical science 的平均GRE分数没有显著差异

(c) $W = q_{\alpha(k, df)} \sqrt{\frac{MSE}{n}}$

$= q_{0.05(3,27)} \sqrt{\frac{8367.78}{10}}$

$= 3.49\ (28.9271)$

$= 100.96$

Life vs Physical $= |608 - 653| = 45 < 100.96$, Not significant difference

Life vs Social $= |608 - 532| = 76 < 100.96$, Not significant difference

Physical vs Social $= |653 - 532| = 121 > 100.96$, significant difference

∴ Physical Sciences 和 Social Sciences significant difference in the average GRE scores

DATA SET DS1123

24. Where to Shop? Do you shop at the grocery store closest to home or do you look for the store that has the best prices? We compared the regular prices at four different grocery stores for eight items purchased on the same day.

Items	Stores: Vons	Ralphs	Stater Bros	WinCo
Salad mix, 360 g bag	3.99	2.79	1.99	1.78
Hillshire Farm® Beef Smoked Sausage, 420 g	4.29	4.29	3.99	2.50
Kellogg's Raisin Bran®, 750 g	4.49	5.49	4.49	3.15
Kraft® Philadelphia® Cream Cheese, 250 g	2.99	3.19	2.79	1.48
Kraft® Ranch Dressing, 450 mL	3.19	3.49	3.49	1.48
Treetop® Apple Juice, 2 L	2.99	3.49	3.49	1.58
Dial® Bar Soap, Gold, 125 g	5.99	6.49	5.79	5.14
Jif® Peanut Butter, Creamy, 850 g	5.15	5.49	4.79	4.34

a. What are the blocks and treatments in this experiment?

b. Do the data provide evidence to indicate that there are significant differences in prices from store to store? Support your answer statistically using the ANOVA printout that follows.

c. Are there significant differences from block to block? Was blocking effective?

ANOVA: Price versus Item, Store

Analysis of Variance for Price

Source	DF	SS	MS	F	P
Store	3	13.192	4.3974	26.52	0.000
Item	7	40.995	5.8564	35.32	0.000
Error	21	3.482	0.1658		
Total	31	57.669			

Model Summary

S	R-sq	R-sq(adj)
0.407181	93.96%	91.09%

(a) Blocks: the eight items

Treatments: the four grocery stores (Vons, Ralphs, Stater Bros, WinCo)

(b) $p = 0.0000 < 0.05$

$\therefore$ Reject H_0, there are significant differences in prices from store to store

(c) $p = 0.0000 < 0.05$

$\therefore$ Yes, there are significant differences from block to block, so blocking is effective.

25. Where to Shop?, continued Refer to Exercise 24. The printout that follows provides the average costs of the selected items for the $k = 4$ stores.

Means

Store	N	Price
Ralphs	8	4.34000
Stater Bros	8	3.85250
Vons	8	4.13500
WinCo	8	2.68125

a. What is the appropriate value of $q_{.05}(k, df)$ for testing for differences among stores?

b. What is the value of $\omega = q_{.05}(k, df)\sqrt{\frac{MSE}{b}}$?

c. Use Tukey's pairwise comparison test among stores used to determine which stores differ significantly in average prices of the selected items.

(a) $q_{0.05}(k, df)$

$= q_{0.05}(4, 21)$

$= 3.96$

(b) $W = q_{0.05}(k, df)\sqrt{\frac{MSE}{N}}$

$= 3.96\sqrt{\frac{0.1458}{8}}$

$= 0.57$

(c) Ralphs vs Stater $= |4.34 - 3.8525| = 0.4875 < 0.57$

Ralphs vs Vons $= |4.34 - 4.135| = 0.205 < 0.57$

Ralphs vs WinCo $= |4.34 - 2.68125| = 1.65875 > 0.57$

Stater vs Vons $= |3.8525 - 4.135| = 0.2825 < 0.57$

Stater vs WinCo $= |3.8525 - 2.68125| = 1.17125 > 0.57$

Vons vs WinCo $= |4.135 - 2.68125| = 1.45375 > 0.57$

$\therefore$ WinCo has significant differences in average prices of the selected items.

Ralphs, Vons, Stater Bros not significant differences

DATA SET DS1124

26. Rocket Propellants An experiment was conducted to compare three mixtures of components in making rocket propellant. Five samples of each mixture were obtained for testing. Five investigators measured the propellant thrust in pounds for each of the mixtures, with the mixtures presented in random order.

	Mixtures		
Investigators	1	2	3
1	42	39	45
2	37	36	40
3	43	48	50
4	41	47	52
5	39	42	47

Mixture 1: $T_1 = 202$, $n_1 = 5$, $\bar{T}_1 = 40.4$

Mixture 2: $T_2 = 212$, $n_2 = 5$, $\bar{T}_2 = 42.4$

Mixture 3: $T_3 = 234$, $n_3 = 5$, $\bar{T}_3 = 46.8$

$X_{..} = 648$, $\bar{X}_{..} = 43.2$

$N = 15$

$b = 5$

$k = 3$

Investigators

$B_1 = 126$, $B_4 = 140$

$B_2 = 113$, $B_5 = 128$

$B_3 = 141$

a. Identify the treatments and the blocks in this experiment and provide an appropriate analysis of variance.

(a) Treatments : Three types of Mixtures

Blocks : Five Investigators

$H_0: \mu_1 = \mu_2 = \mu_3$

$H_a: \mu_1, \mu_2, \mu_3$ 不完全相等

$H_0: \mu_{B1} = \mu_{B2} = \mu_{B3} = \mu_{B4} = \mu_{B5}$

$H_a: \mu_{B1}, \mu_{B2}, \mu_{B3}, \mu_{B4}, \mu_{B5}$ 不完全相等

ANOVA

Sources	SS	df	MS	F
Treatments	107.2	$k-1 = 2$	$\frac{107.2}{2} = 53.6$	$\frac{53.6}{4.85} = 11.05$
Blocks	176.4	$b-1 = 4$	$\frac{176.4}{4} = 44.1$	$\frac{44.1}{4.85} = 9.09$
Error	38.8	$(k-1)(b-1) = 8$	$\frac{38.8}{8} = 4.85$	
Total	322.4	$N-1 = 14$		

$$CM = \frac{X_{..}^2}{N} = \frac{648^2}{15} = 27993.6$$

$$SSTR = \frac{\sum T_i^2}{b} - CM = \frac{202^2}{5} + \frac{212^2}{5} + \frac{234^2}{5} - 27993.6 = 107.2$$

$$SSTO = \sum X_{ij}^2 - CM = 28316 - 27993.6 = 322.4$$

$$SSB = \frac{\sum B_i^2}{k} - CF = \frac{126^2}{3} + \frac{113^2}{3} + \frac{141^2}{3} + \frac{140^2}{3} + \frac{128^2}{3} - 27993.6 = 28170 - 27993.6 = 176.4$$

$$SSE = SSTO - SSTR - SSB = 38.8$$

b. Is there sufficient evidence to conclude there is a significant difference in the propellant thrust for the mixtures? Use $\alpha = .01$.

(b) $F = 11.05$

$F_{2,8,0.01} = 8.6491$

$\because 11.05 > 8.6491$

$\therefore$ Reject H_0

c. Was blocking effective? Use the p-value to draw your conclusions and justify your answer.

(c) $F = 9.09$, $df_1 = 4$, $df_2 = 8$

$7.0 < 9.09 < 9.6$

$0.005 < p < 0.01$

$\because p < 0.01$

$\therefore$ Reject H_0

d. Use Tukey's pairwise comparisons with $\alpha = .01$ to determine which propellant mixtures differed from others. Summarize your results.

$$w = q_{0.01}(3,8)\sqrt{\frac{4.85}{5}}$$
$$= 5.64\sqrt{\frac{4.85}{5}}$$
$$= 5.55$$

Mix 1 vs Mix 2 $= |40.4 - 42.4| = 2.0 < 5.55$, Not significant difference

Mix 1 vs Mix 3 $= |40.4 - 46.8| = 6.4 > 5.55$, significant difference

Mix 2 vs Mix 3 $= |42.4 - 46.8| = 4.4 < 5.55$ Not significant difference